Communication-Adaptive CBFs for Multi-Agent Navigation – Research Statement

Motivation and Problem Statement: Teams of robots are increasingly deployed in disaster response scenarios — collapsed buildings, subterranean sites, and wildfire zones — with rapidly changing, communication-limited conditions [1]. Communication failures degrade coordination, causing redundant coverage, incomplete mapping, and unsafe behaviors. Because lives may depend on rapid, accurate search, **ensuring reliable coordination under imperfect communication is mission-critical for saving lives**. Existing multi-robot methods handle communication loss in various ways, from reactive recovery [2] to conservative connectivity maintenance [3], yet few explicitly incorporate *predicted* communication quality into safety-critical control. This leaves a critical gap: *maintaining safety constraints (collision avoidance, connectivity) under unreliable communication*. My research addresses this challenge with **Communication-Adaptive Control Barrier Functions (CA-CBFs)**, which predict communication quality and adjust safety constraints and paths in real time. The framework enables multi-agent teams to explore efficiently, coordinate safely despite intermittent or noisy communication, and adapt to unknown environments. While motivated by disaster response, CA-CBFs can also apply to remote terrestrial and space exploration.

Background & Research Gap: Current multi-robot methods typically treat safety and communication-based path planning separately. Control Barrier Functions (CBFs) ensure robots stay within safe boundaries by defining a *barrier function* $h(x)$ with $h(x) \geq 0$ indicating safe states. Many CBF-based multi-robot methods assume reliable communication or enforce fixed connectivity [3, 4], yet real environments feature intermittent signals that affect coordination capabilities. While some CBF variants adapt to time-varying conditions [5, 6], they do not adapt safety constraints based on *predicted* communication degradation. Additionally, communication-aware planners predict signal quality [7, 8] but lack real-time safety guarantees. **CA-CBFs bridge this gap by embedding predictive communication models directly into the barrier function framework.** Each robot can proactively adjust its behavior and coordinate with teammates, maintaining safe operation even when connectivity is uncertain or asymmetric. This approach unifies predictive communication modeling and real-time, safety-critical control for multi-agent systems.

Proposed Solution & Research Plan: My research will develop Communication-Adaptive Control Barrier Functions (CA-CBFs), a novel framework where multi-agent safety constraints adjust to predicted communication quality $c(t) \in [0, 1]$ — representing signal strength, bandwidth, and connectivity.

The key innovation is the barrier function $h(x, c) = d_{\max}(c)^2 - d(\text{robot}_i, \text{robot}_j)^2$, where the maximum allowable separation $d_{\max}(c)$ between robots i and j grows when communication is strong ($c(t) \approx 1$, enabling aggressive exploration) and shrinks as it degrades ($c(t) \approx 0$, for conservative coordination), as shown in Figure 1. Each agent predicts $c(t + \Delta t)$ from path and terrain maps, and consensus aligns communication estimates despite sensor noise. Model-predictive planning uses predicted communication quality, and a QP solver minimally adjusts trajectories to satisfy the CBF constraint, enabling proactive adaptation before communication loss.

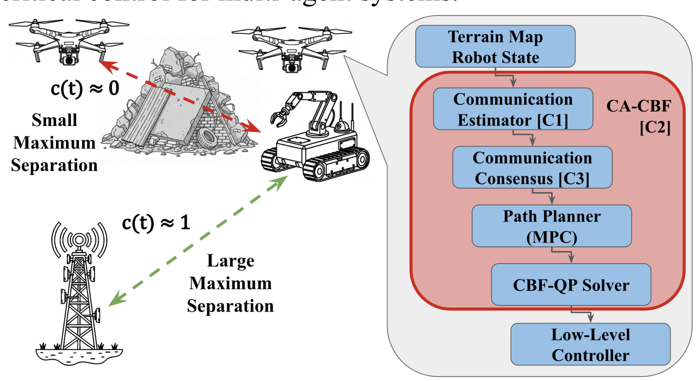

Figure 1: CA-CBF concept for disaster response. Communication quality $c(t)$ adapts separation limits. Predicted $c(t)$ informs planning, and CA-CBFs minimally adjust controls for safety. The three core challenges (C1, C2, C3) are labeled in context.

This framework addresses three core challenges — [C1] computationally-efficient predictive communication models for real-time operation, [C2] CA-CBFs that ensure safety under degraded communication, and [C3] asymmetric beliefs about communication quality between robots — via four objectives:

Objective 1: Develop a Predictive Communication Model for Multi-Robot Systems I will develop physics-based models to predict communication quality $c(t+\Delta t)$ from robot trajectories and available environmental information (building blueprints, satellite imagery, or incrementally-built SLAM maps), leveraging occlusion reasoning [9] and path planning [7] for real-time forecasting. Validation will compare against experimentally-measured ground-truth signals in both simulation and hardware testbeds [Timeline: Years 1-2; C1].

Objective 2: Formulate Communication-Adaptive Control Barrier Functions I will formulate CA-CBFs by designing communication-parameterized barrier functions $h(x, c(t))$ and utilize already-existing QP solvers for real-time control. The formulation will define how $d_{\max}(c)$ adapts to communication quality while maintaining optimization convexity. Initial two-robot demonstrations will verify coordination constraints: maintaining connectivity and avoiding collisions [Timeline: Years 1-2; C2].

Objective 3: Extend CA-CBFs for Scalable, Distributed Coordination I will develop distributed consensus protocols enabling robots to align communication quality estimates despite sensor noise and asymmetry. Robots will exchange compressed estimates using average consensus or gossip algorithms [10] to reach agreement on local communication conditions, enabling decentralized safety decisions. Validation will demonstrate safe operation in teams of 5 to 10 agents [Timeline: Years 2-3; C3].

Objective 4: Validate CA-CBFs in Simulated and Real-World Multi-Robot Missions I will demonstrate CA-CBFs in mock mission scenarios (disaster response, lunar exploration) with Monte Carlo simulations to statistically validate safety guarantees and hardware testing on ground robots and aerial drones equipped with ranging sensors and wireless communication modules. Performance will be benchmarked against baseline methods (fixed-connectivity constraints, reactive methods, standard multi-agent CBFs) using metrics including collision rate, area coverage per unit time, and network connectivity maintenance. The outcome will be an open-source CA-CBF library for the robotics community to use [Timeline: Years 3-4; C2 and C3].

Intellectual Merit:

My research improves safe multi-robot coordination under unreliable communication. Existing mission-critical approaches do not adapt constraints based on predicted communication quality. I propose Communication-Adaptive Control Barrier Functions (CA-CBFs), which adjust coordination constraints (separation, collision avoidance) in real time using predicted communication. Contributions include adaptive barrier function design with communication prediction [C1], a unified framework linking communication-aware planning to formal safety [C2], and hardware tests showing robustness under asymmetric communication [C3].

Broader Impacts:

Societal Impact and Partnerships: While volunteering in Houston after Hurricane Harvey, I witnessed the need for timely aid in disaster zones. By maintaining safe coordination under degraded communication, CA-CBFs improve robot team reliability and timely disaster response. I will explore pathways to transfer this technology to emergency management agencies and first responders, and seek collaboration with NASA JPL through my graduate institution for applications in extraterrestrial exploration.

Open-Source Materials and STEM Education: I will release an open-source ROS 2 implementation of CA-CBFs on GitHub with beginner-friendly tutorials and modular examples designed to lower barriers for students new to robotics, regardless of institutional resources. I will maintain this repository throughout my PhD, responding to user issues and incorporating feedback to improve usability. Building on my experience with the Mark Rober video that reached 75 million views and my TA experience, I will create 3-4 educational videos or interactive demonstrations about robot safety for high school students and college undergraduates on YouTube. Each YouTube video will use real robot footage and visual analogies (e.g., "robot traffic laws" for barrier functions) to make abstract safety concepts intuitive, before diving into introductory theory. These resources will make safe autonomy concepts accessible to both researchers and students.

Mentoring and Outreach: I will mentor 2–3 undergraduate researchers during my PhD, providing structured projects and supporting their presentations at university symposiums. Building on my experience growing a research team from 3 to over 20 members, I will develop clear onboarding materials that other graduate students can use for their mentees. Each year, I will participate in 1–2 outreach programs — such as elementary school visits — bringing interactive demonstrations and short lessons on how robots sense and act, similar to BYU's Lab-o-ween as mentioned in my personal statement. I will collect short surveys from students and teachers to assess engagement and understanding, and use this feedback to refine future activities. I will document these efforts in my annual NSF reports to ensure continual improvement and lasting impact.

References [1] Wilk-Jakubowski et al. *Technology in Society*, 2022; [2] Fiasché et al. *IFAC*, 2019; [3] Capelli et al. *ICRA*, 2021; [4] Gonçalves et al. *Robotics and Autonomous Systems*, 2024; [5] Taylor et al. *IEEE TAC*, 2020; [6] Gao et al. *arXiv preprint*, 2023; [7] Liu et al. *ACC*, 2017; [8] Wang et al. *Autonomous Robots*, 2023; [9] Staudinger et al. *Acta Astronautica*, 2023; [10] Olfati-Saber et al. *Proc. IEEE*, 2007;